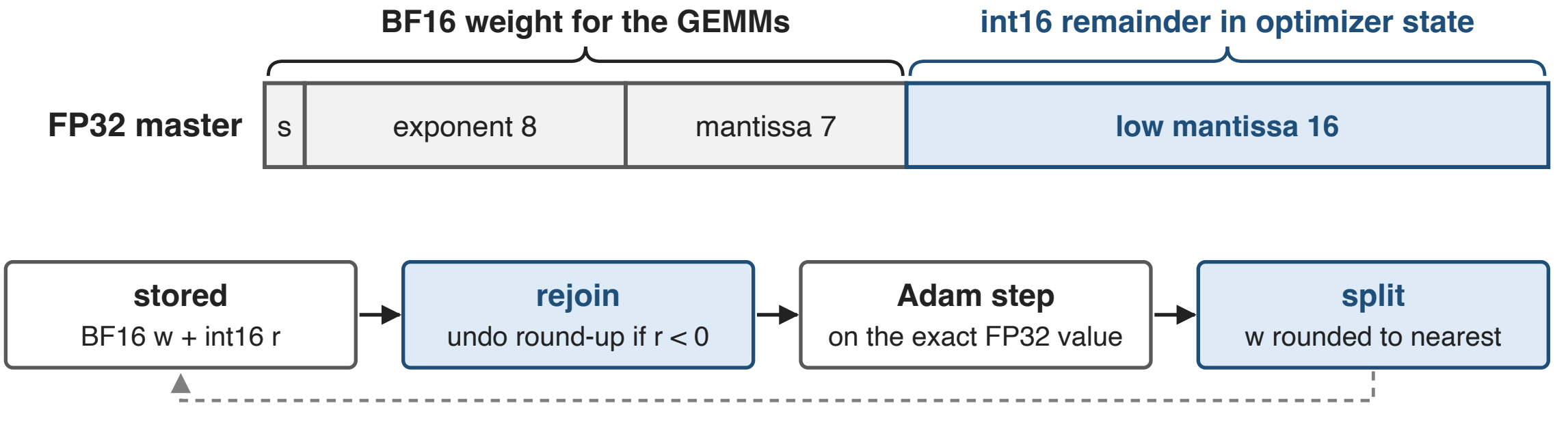
BF16 weight for the GEMMs
int16 remainder in optimizer state
FP32 master
s
exponent 8
mantissa 7
low mantissa 16
stored
BF16 w + int16 r
rejoin
undo round-up if r < 0
Adam step
on the exact FP32 value
split
w rounded to nearest